BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Strings and Input Checking

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is the result of concatenating 'AB' and '12'?

2. How is an item accessed in a 1D array?

A) Using its font
B) Using a file extension
C) Using a network address
D) Using an index

3. "Testing a few inputs proves a program can never contain any fault." Correct this. Explain why.

4. A programmer tests input code while building a booking program, then tests the finished program. Correct both claims. Explain why.
(a) Iterative testing starts only after delivery. (b) Final testing removes the need for earlier tests.

Today's objectives

Manipulate strings and check inputs against rules. Use the technique in an independent scenario.

Key vocabulary - Read, cover, recall

Length check
A string's .length property can check that a minimum number of characters were supplied.

Validation
Checking entered data against rules improves reliability but does not verify identity.

Input checking
Reject a value that fails a stated rule and request another valid one.

Authentication
Checking a user's identity is different from checking whether input fits a format.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Length check	A string's .length property can check that a minimum number of characters were supplied. Apply this in a school portal username.
Validation	Checking entered data against rules improves reliability but does not verify identity. Apply this in a library account.
Input checking	Reject a value that fails a stated rule and request another valid one. Apply this in a club membership.
Authentication	Checking a user's identity is different from checking whether input fits a format. Apply this in a mobile app sign-up.

I Do - One worked example

A school portal username is being developed. Explain how length check and validation are used. [4 marks]

1. Read the scenario and identify the data the program needs.

2. Length check [1]; A string's .length property can check that a minimum number of characters were supplied. [1].

3. Validation [1]; Checking entered data against rules improves reliability but does not verify identity. [1].

4. Two linked points for each concept.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A library account uses length check. Explain how it helps. [2 marks]

2. Explain validation in a club membership. [2 marks]

3. A mobile app sign-up uses input checking and authentication. Explain both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Replace the fixed password with an input instruction.

```
password = "abc123"
if password.length >= 6 then
print("Length OK")
else
print("Too short")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State TWO relevant techniques from this topic. [2 marks]

Q3. Explain how validation and input checking help with a video service username. [4 marks]

Q4. A programmer designs a sports team password. Explain why length check, input checking and authentication matter. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A string's .length property can check that a minimum number of characters were supplied. ☐ T ☐ F
2. A string's .length property can check that a minimum number of characters were supplied ☐ T ☐ F
3. Checking entered data against rules improves reliability but does not verify identity ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 AB12, with no automatically inserted space. ☐

Q2 D - Using an index ☐

Q3 To identify faults and check behaviour against requirements using expected results. ☐

Q4 (a) Iterative testing checks and fixes parts while building the program. (b) Final testing checks the finished program against its requirements. Both are needed. ☐

Code First - Model

```
password = input("Test password:")
if password.length >= 6 then
print("Length OK")
else
print("Too short")
endif
```

Output: Length OK (if student enters abc123)

We Do - Check the model

1. Length check [1]; A string's .length property can check that a minimum number of characters were supplied. [1]. ☐

2. Validation [1]; Checking entered data against rules improves reliability but does not verify identity. [1]. ☐

3. Input checking [1]; Reject a value that fails a stated rule and request another valid one. [1]. Authentication [1]; Checking a user's identity is different from checking whether input fits a format. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Length check [1]; Input checking [1]. ☐

Q3 Validation [1]; Checking entered data against rules improves reliability but does not verify identity. [1]. Input checking [1]; Reject a value that fails a stated rule and request another valid one. [1]. ☐

Q4 Length check [1]; A string's .length property can check that a minimum number of characters were supplied. [1]. Input checking [1]; Reject a value that fails a stated rule and request another valid one. [1]. Authentication [1]; Checking a user's identity is different from checking whether input fits a format. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: